DHCP + ROAS Configuration

Lab Documentation & Configuration Evidence

Lab Objective	Demonstrate DHCP address assignment, VLAN-based addressing, Router-on-a-Stick configuration, and a router acting as a DHCP client.
Addressing	VLAN 10 → 1.0.0.0/24 | VLAN 20 → 2.0.0.0/24 | R2 link → 10.0.0.0/24
Evidence	Topology, PC DHCP results, VLAN 20 DHCP/ROAS configuration, and R2 DHCP client verification.

Completed lab evidence compiled from the submitted screenshots.

1. Lab Topology

The completed topology contains a DHCP-capable router/server connected to R2 over the 10.0.0.0/24 network. The same server-side device is connected to R11, which provides the VLAN-facing routing function. VLAN 10 contains PC3 and PC4 using the 1.0.0.0/24 network, while VLAN 20 contains PC5 and PC6 using the 2.0.0.0/24 network.

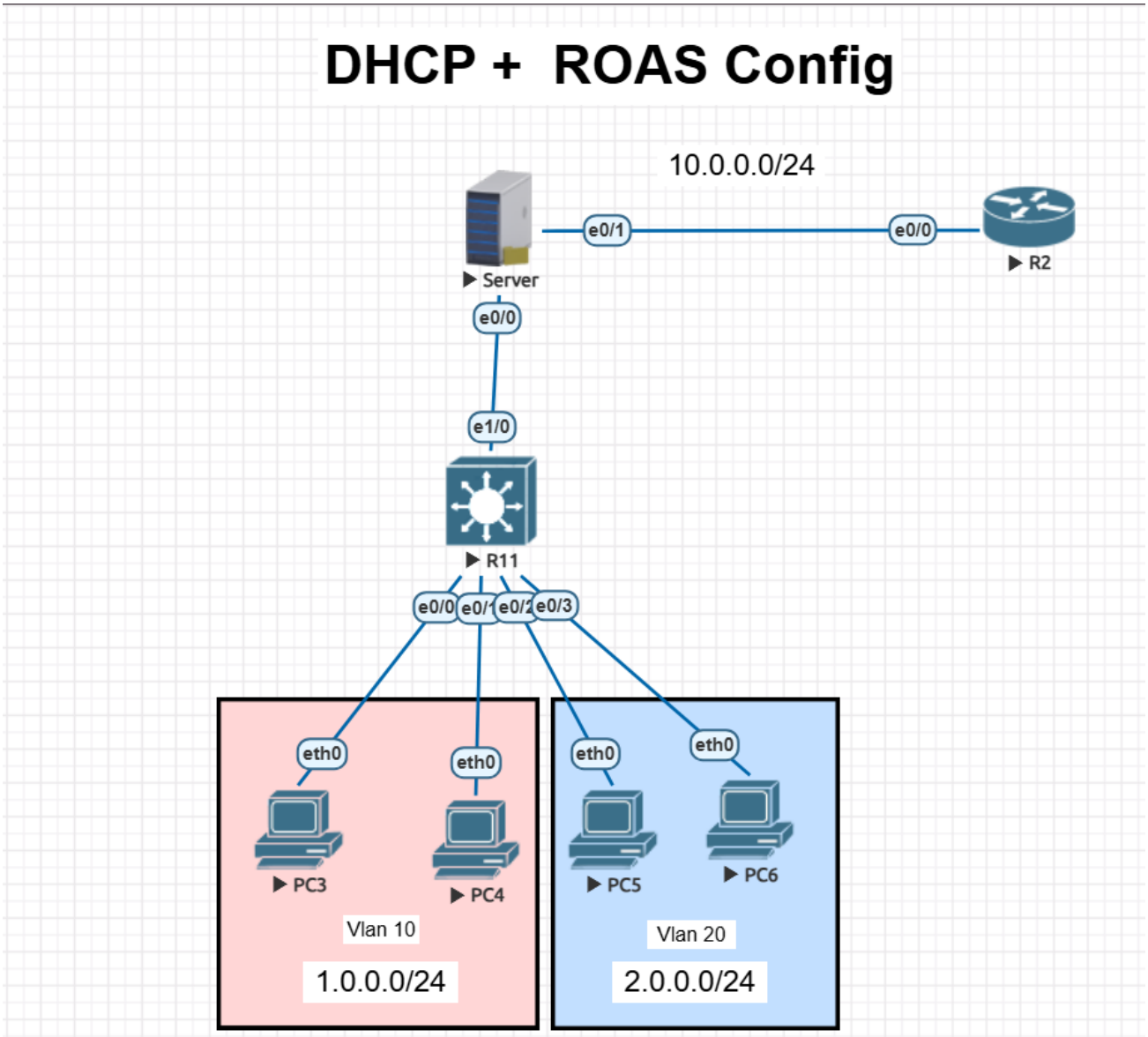


Figure 1 — Final documented topology: DHCP + ROAS configuration.

Segment	Devices	Network	Observed Gateway
VLAN 10	PC3, PC4	1.0.0.0/24	1.0.0.254
VLAN 20	PC5, PC6	2.0.0.0/24	2.0.0.254
R2 DHCP Link	R2 e0/0	10.0.0.0/24	DHCP assigned

2. VLAN 10 — DHCP Verification

PC3 and PC4 were configured as DHCP clients in VLAN 10. The verification screens show successful address acquisition from the 1.0.0.0/24 network. The observed default gateway is 1.0.0.254, confirming the VLAN's gateway configuration.

PC3 — DHCP lease verification (1.0.0.2/24)

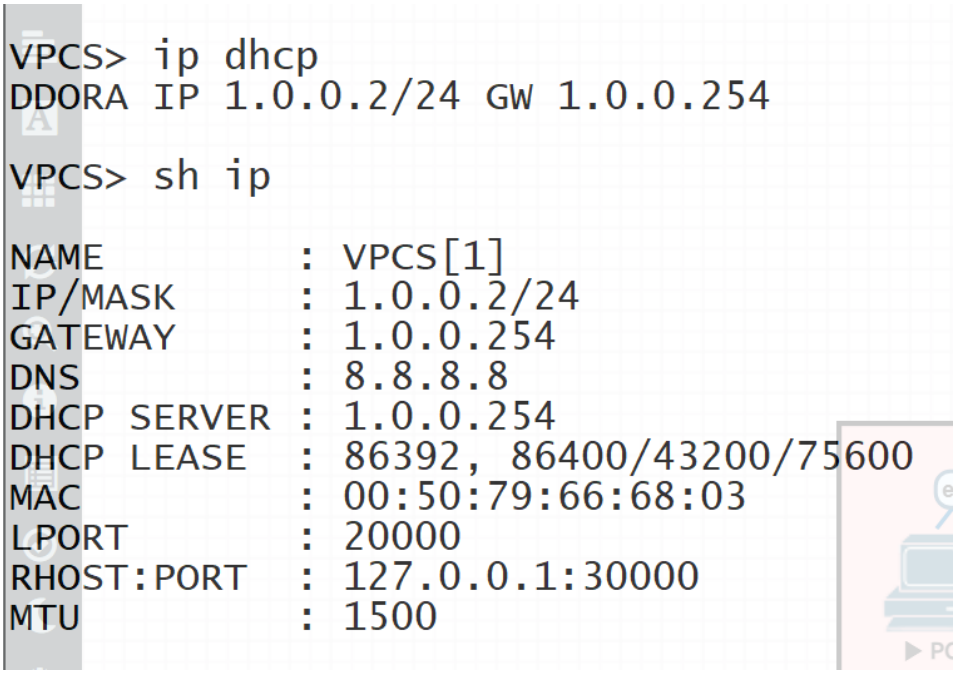


Figure — PC3 successfully received an address through DHCP; gateway shown as 1.0.0.254.

PC4 — DHCP lease verification (1.0.0.1/24)

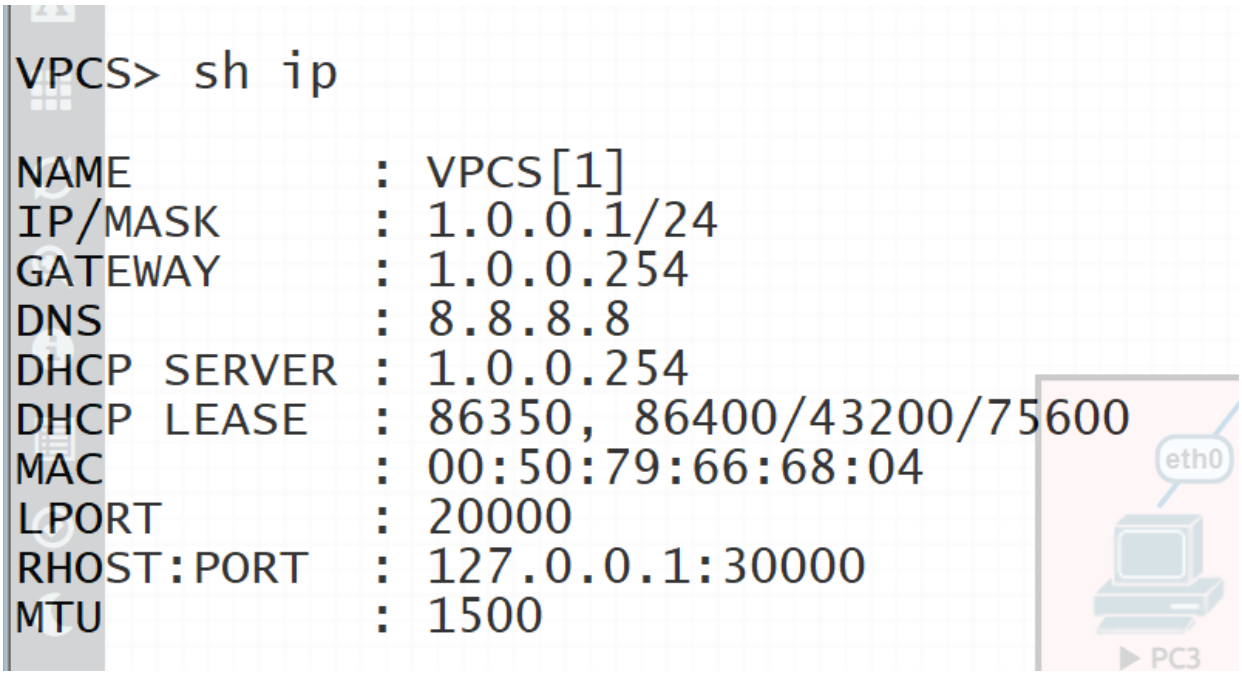


Figure — PC4 successfully received an address through DHCP; gateway shown as 1.0.0.254.

3. VLAN 20 — DHCP Verification

PC5 and PC6 were configured as DHCP clients in VLAN 20. Both verification screens show successful DHCP assignment from the 2.0.0.0/24 network, with 2.0.0.254 recorded as the default gateway and DHCP server address.

PC5 — DHCP lease verification (2.0.0.2/24)

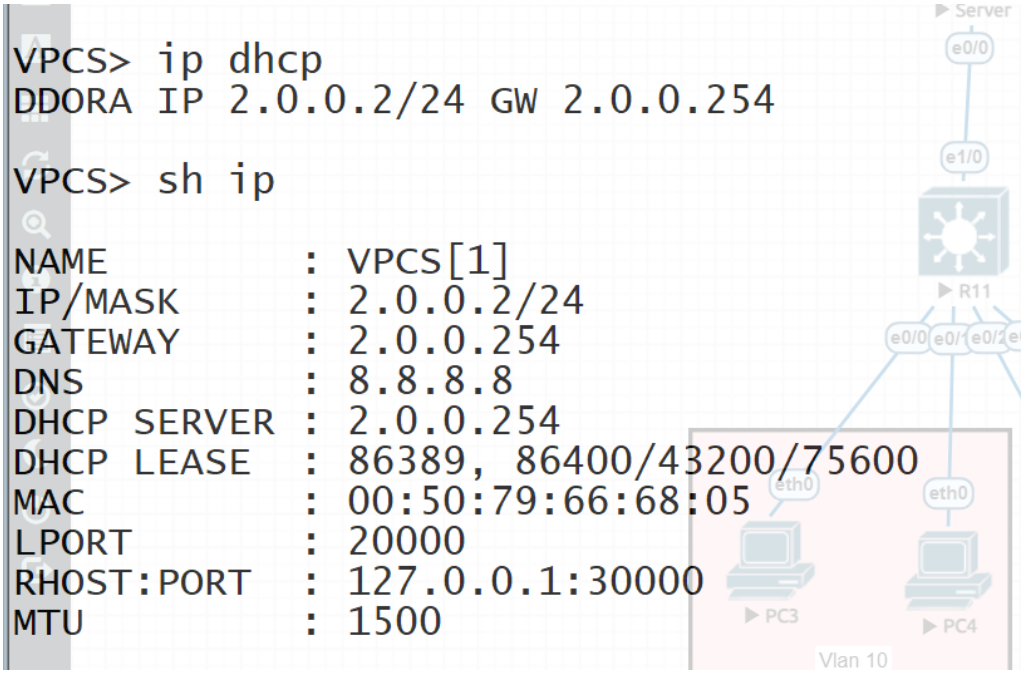


Figure — PC5 successfully received an address through DHCP; gateway shown as 2.0.0.254.

PC6 — DHCP lease verification (2.0.0.1/24)

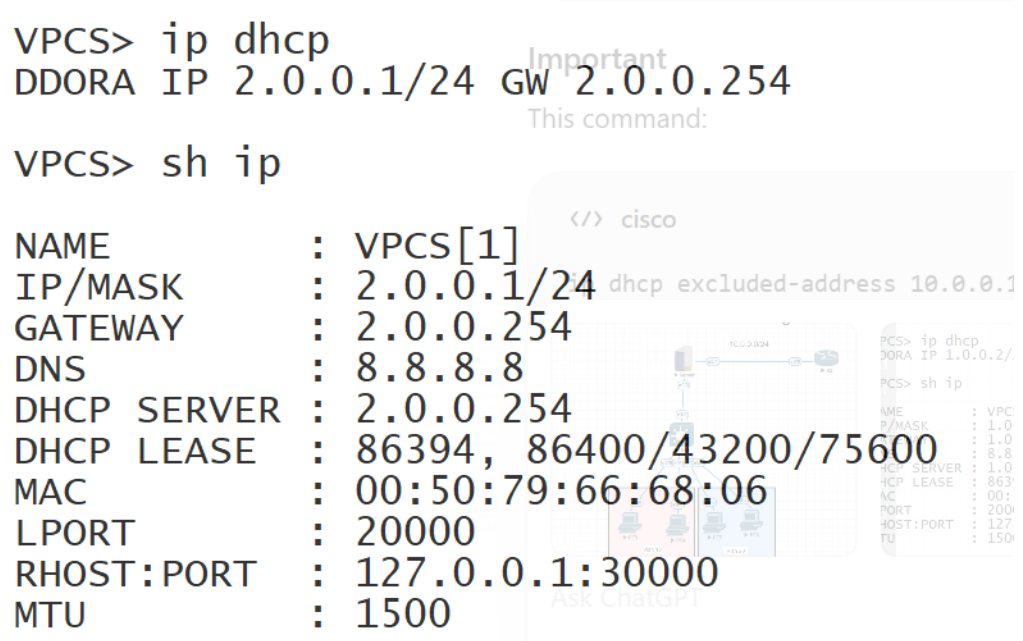


Figure — PC6 successfully received an address through DHCP; gateway shown as 2.0.0.254.

4. ROAS / VLAN 20 Configuration Evidence

The configuration evidence shows a VLAN 20 subinterface created on the router-facing interface, with 802.1Q tagging and the VLAN 20 gateway address. This is the routing component that allows VLAN 20 hosts to use the router as their Layer-3 gateway.

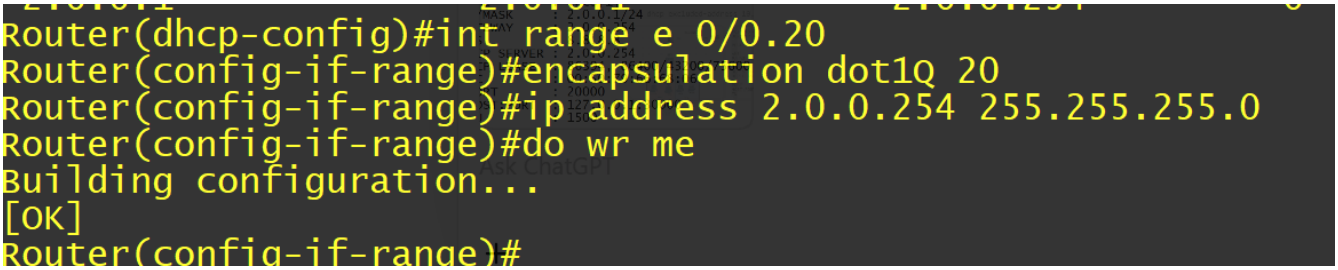


Figure — VLAN 20 subinterface and 802.1Q configuration.

DHCP pool evidence

The DHCP configuration screen shows the VLAN 20 pool, network statement, default gateway, DNS value, and the excluded gateway address.

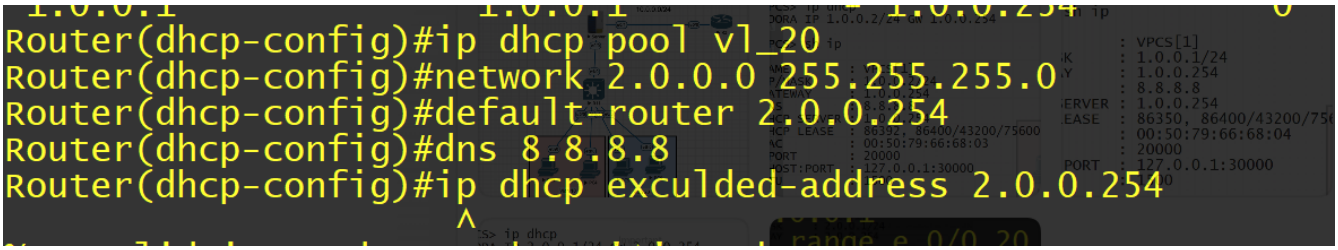
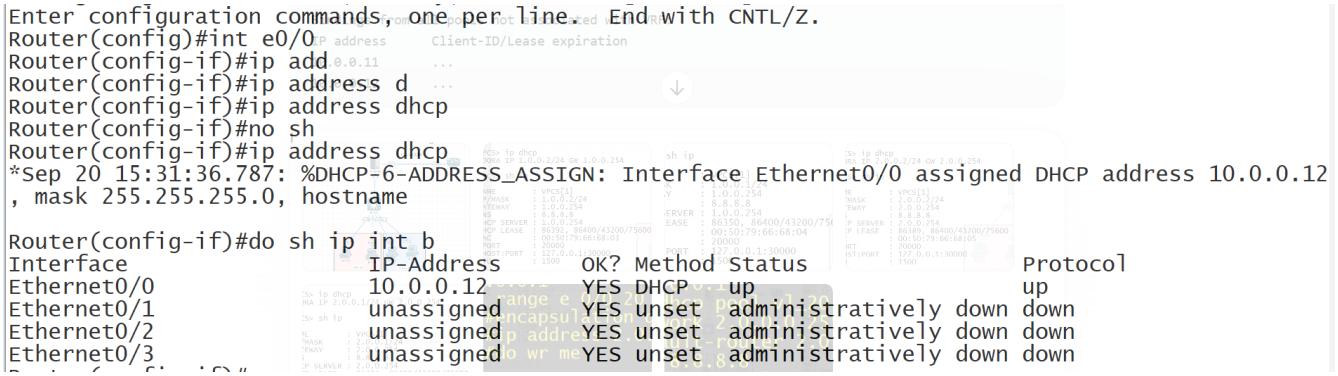


Figure — VLAN 20 DHCP pool configuration evidence.

5. R2 — DHCP Client Verification

R2 was configured as the DHCP client on Ethernet0/0. The verification output confirms that the interface successfully received 10.0.0.12/24 dynamically. The interface is operational (up/up), demonstrating successful DHCP communication over the 10.0.0.0/24 link.



```
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int e0/0
Router(config-if)#ip add
Router(config-if)#ip address d
Router(config-if)#ip address dhcp
Router(config-if)#no sh
Router(config-if)#ip address dhcp
*Sep 20 15:31:36.787: %DHCP-6-ADDRESS_ASSIGN: Interface Ethernet0/0 assigned DHCP address 10.0.0.12
, mask 255.255.255.0, hostname

Router(config-if)#do sh ip int b
Interface                  IP-Address      OK? Method Status                Protocol
Ethernet0/0                10.0.0.12       YES DHCP   up                    up
Ethernet0/1                unassigned      YES unset  administratively down down
Ethernet0/2                unassigned      YES unset  administratively down down
Ethernet0/3                unassigned      YES unset  administratively down down
```

Figure — R2 Ethernet0/0 receiving 10.0.0.12 through DHCP.

Basic Commands Used

```
interface e0/0.20
encapsulation dot1Q 20
ip address 2.0.0.254 255.255.255.0
ip dhcp pool vl_20
network 2.0.0.0 255.255.255.0
default-router 2.0.0.254
dns 8.8.8.8
ip dhcp excluded-address 2.0.0.254
ip address dhcp
no shutdown
do show ip interface brief
```

Result summary: DHCP address assignment was demonstrated for VLAN 10 and VLAN 20 hosts, and R2 was verified as a DHCP client on the 10.0.0.0/24 segment.